

UNIFORM CIRCULAR MOTION (01)

Aim

- A. Verification of the acceleration of an object in a uniform circular motion, $a = \frac{v^2}{r}$
- B. Determination of the force constant of a given rubber band.

Materials and Apparatus

Experimental setup, string, pulley, masses with a hanger, rubber bands, stopwatch, clamp, digital balance, and pointer.

Theory

The speed of an object traveling in a circle (v) is given by,

$$v = \frac{2\pi r}{T} \quad (01)$$

where,

r : radius of the circular path

T : period of rotation

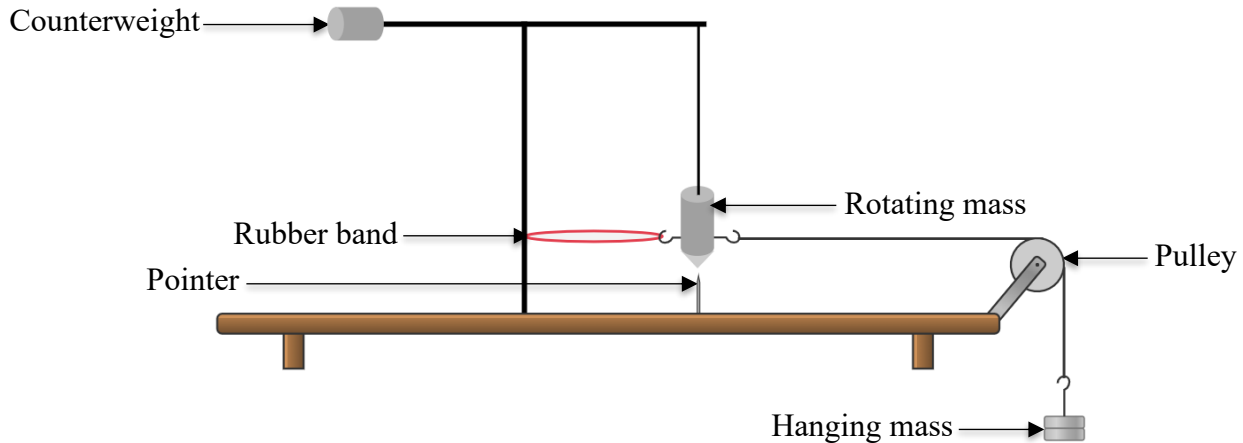


Fig. 1: The schematic diagram of the experimental setup

For an object going around in a circle at a constant speed, the direction of its velocity is changing. So, there is an acceleration towards the center of the circle, given by:

$$a = \frac{v^2}{r} \quad (02)$$

Part A

In order to verify equation (2),

Assume that,

$$a = \frac{v^\alpha}{r^\beta} \quad (03)$$

where, α and β are integers

If r is constant,

$$\ln(a) = \alpha \ln(v) - \beta \ln(r) \quad (04)$$

When the elastic force of the rubber band is balanced by the hanging mass (m),

$$a = \left(\frac{m}{M} \right) g \quad (05)$$

where,

m : hanging mass

M : rotating mass

By obtaining any two data set,

$$\ln(a_1) = \alpha \ln(v_1) - \beta \ln(r) \quad (06)$$

$$\ln(a_2) = \alpha \ln(v_2) - \beta \ln(r) \quad (07)$$

From equations (06) and (07),

$$\ln(a_1) - \ln(a_2) = \alpha [\ln(v_1) - \ln(v_2)] \quad (08)$$

$$\ln\left(\frac{a_1}{a_2}\right) = \alpha \ln\left(\frac{v_1}{v_2}\right) \quad (09)$$

$$\ln\left(\frac{m_1}{m_2}\right) = \alpha \ln\left(\frac{T_2}{T_1}\right) \quad (10)$$

$$\alpha = \frac{\ln\left(\frac{m_1}{m_2}\right)}{\ln\left(\frac{T_2}{T_1}\right)} \quad (11)$$

From equation (06),

$$\beta = \frac{\alpha \ln(v_1) - \ln(a_1)}{\ln(r)} \quad (12)$$

Part B

When the rotating mass (M) is in a uniform motion, the centripetal force (F_c) acting on mass M is given by,

$$F_c = M \left(\frac{v^2}{r} \right) \quad (13)$$

The tension of the rubber band is balanced by the centripetal force as it is connected to the mass in a circular motion.

If the extension of the rubber band is x ,

$$M \left(\frac{v^2}{r} \right) = kx \quad (14)$$

where, k is the force constant of the rubber band

Since,

$$x = r - l_0 \quad (15)$$

where, l_0 is the natural length of the rubber band

$$M \left(\frac{v^2}{r} \right) = k(r - l_0) \quad (16)$$

$$\frac{v^2}{r} = \left(\frac{k}{M} \right) r - \frac{kl_0}{M} \quad (17)$$

$\downarrow \qquad \qquad \downarrow \quad \downarrow \quad \downarrow$
 $y \qquad = \qquad m \quad x \quad - \quad c$

The force constant k can be determined from the graph of $\left(\frac{v^2}{r} \right)$ vs r .

Procedure**Part A**

1. Set the radius to a suitable value and place the reference pin under the mass when the hanging string is vertical.

Q: *How do you determine the suitable value for the radius to start the experiment?*

2. Connect a rubber band to the mass and the fixed rod.
3. Connect a string to the mass so that the string goes over the pulley. At the other end of the string, attach a weight hanger. Add masses to the weight hanger until the mass is aligned with the reference pin.
4. Remove the mass hanger and the relevant string and rotate the system until the mass is aligned with the reference pin.
5. Measure the time for a suitable number of revolutions (while keeping the rotating speed constant).

Q: *How do you decide the number of revolutions to minimize the error associated with the period?*

6. Repeat the above procedure for a different number of rubber bands.

Part B

1. Set the radius to a suitable value and place the reference pin under the mass when the hanging string is vertical.
2. Connect a rubber band to the mass and rotate the system until the mass is aligned with the reference pin.
3. Measure the time for a suitable number of revolutions (while keeping the rotating speed constant).
4. Repeat the above procedure for different values of radius.

Q: *What are the other relevant measurements to be taken to verify the relationship and to calculate k ?*